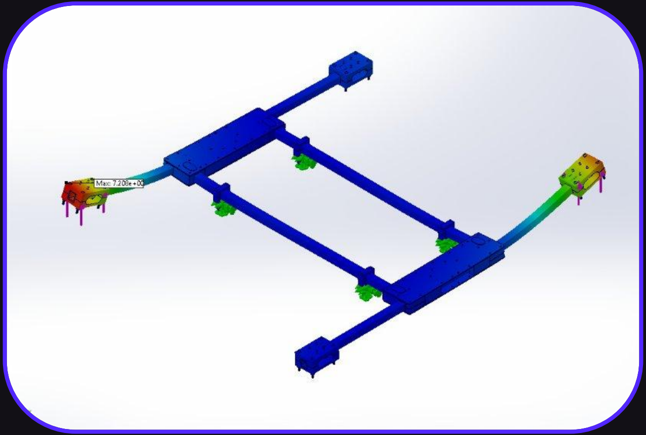
The DRXC Team

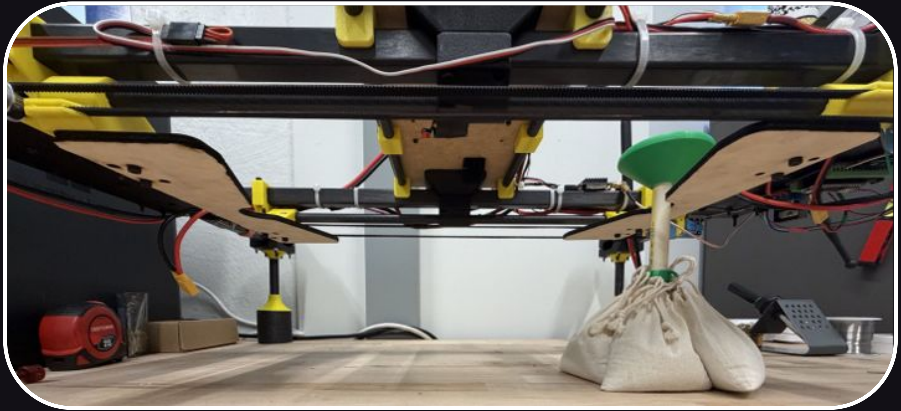
DRXC is the Drones, Radio Control, Experimental Aircraft Club at Georgia Tech. As Chief Engineer, I led the 6 different subteams responsible for the design and implementation of the vehicle.

The team won first place at the 2026 DBVF flyoff, and was invited to present our aircraft at VFS's Forum 82 in Florida.

Vehicle Optimization

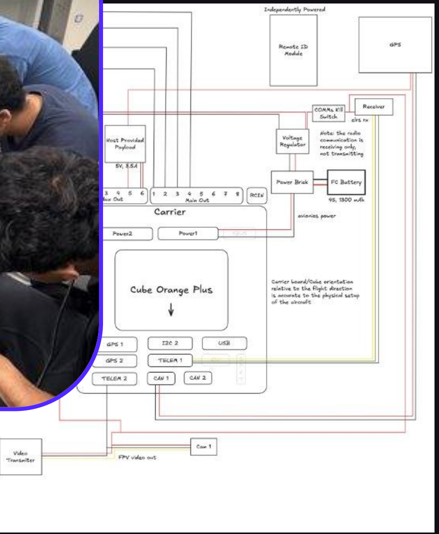
To ensure flyoff performance, Finite Element Analysis (FEA) was conducted under worst-case motor loads, optimizing the airframe and reducing vehicle weight by 20%. Testing the payload system across all possible pickup orientations, as shown below, allowed errors to be identified, addressed, and resolved.





Electronics Hardware

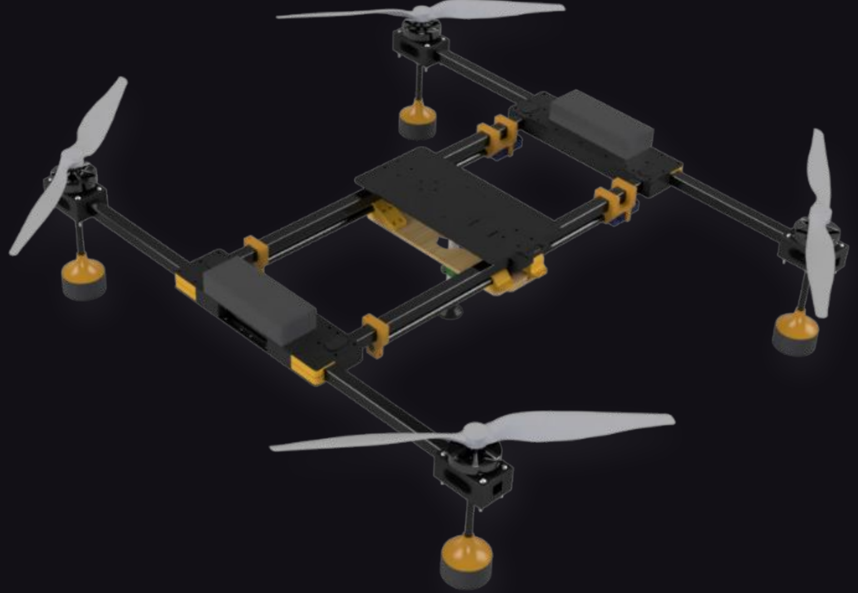
The vehicle utilized a Cube Orange flight controller and ran off of Ardupoiot Mission Planner. Two 12V 3s batteries powered the aircraft's flight system. A separate microcontroller was used to control the pickup and drop mechanism.



The ground station controls the connection to the vehicle, including an RTK GPS antenna, as well as the flight camera monitoring system, all while having a 5 minute setup process.

Designing the Vehicle

The quadcopter configuration was chosen due to plenty of documentation from previous years. The H-frame design allowed the vehicle to remain stiff while also having a large pickup area for the challenge.



The payload system saw many iterations, but the highlight throughout our design was the linear actuation of the system. Small DC motors ran a hub down the length of a carbon rod through a gantry style belt system enabling a symmetric close.